ANANYA SRIDHAR

AI Engineer — Generative AI, Speech AI, Computer Vision & Production ML Systems

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Languages: English, Hindi, Tamil

PROFESSIONAL SUMMARY

Results-driven AI Engineer with hands-on production experience architecting, building, and deploying scalable Generative AI, Speech AI, and Computer Vision systems. Proven track record deploying full-stack AI applications serving 10,000+ financial advisors (Nivesh AI) and clinical speech therapy platforms across 8 Indian languages (Manas Learning). Skilled in low-latency REST APIs and microservices (FastAPI, WebSockets, Docker), Agentic workflows (LangGraph, RAG, Vector DBs), and cloud deployments on Azure with automated CI/CD pipelines.

TECHNICAL SKILLS

- **Languages:** Python (Proficient), C++, SQL, TypeScript, JavaScript, Bash
- **Generative AI & LLMs:** LangGraph, LangChain, RAG (Retrieval-Augmented Generation), Prompt Engineering, Vector Databases (ChromaDB, pgvector), LLM Evaluation, Hugging Face Transformers, PyTorch, LoRA/PEFT
- **Speech AI & Audio Processing:** Faster-Whisper, OpenAI Whisper, Vosk, Coqui TTS, Kokoro ONNX, Audio Signal Processing, Librosa, Epitran (G2P), Real-Time Phoneme Verification, ONNX Runtime
- **Computer Vision & Robotics:** OpenCV, MediaPipe Tasks, Vision Transformers (DINOv2), Object Detection & Tracking, Edge AI, MuJoCo 3D Physics Simulation, Reinforcement Learning (PPO, Q-Learning), Multimodal Sensor Fusion, Three.js
- **Cloud, Backend & MLOps:** FastAPI, REST APIs, Microservices, WebSockets, Azure Cloud, Docker, PostgreSQL, Redis, GitHub Actions CI/CD, Linux, Uvicorn, System Design, Unit Testing

PROFESSIONAL EXPERIENCE

AI/ML ENGINEER INTERN (FULL-STACK & SPEECH AI) | MANAS LEARNING

Mar 2026 – Present | Remote / Mumbai, India

- **Platform Architecture & Delivery:** Architected and deployed full-stack pediatric speech therapy platforms (vaaksudhi.manaslearning.com and vaakify.manaslearning.com) delivering clinical assessments across 8 Indian languages.
- **Real-Time Speech Engine:** Built an end-to-end pronunciation scoring pipeline chaining Faster-Whisper, Vosk, and Kokoro/Coqui TTS with phoneme-level acoustic verification (Librosa, Epitran G2P) for instant articulatory feedback.
- **Latency & Inference Optimization:** Optimized model inference pipelines using ONNX Runtime and streaming audio buffers, slashing end-to-end round-trip latency to <300ms.
- **Agentic AI Diagnostic Workflows:** Implemented LangGraph multi-agent diagnostic pipelines coordinating 4 clinical assessment modules to analyze speech errors and auto-generate structured clinician reports.
- **Adaptive Reinforcement Learning:** Formulated a Q-learning difficulty controller that dynamically personalizes therapy exercises based on rolling acoustic accuracy and response latency.
- **Production REST APIs & Database Design:** Engineered high-performance REST APIs and WebSocket endpoints using FastAPI, Uvicorn, and PostgreSQL, implementing Alembic migrations and database connection pooling.
- **Security, Compliance & CI/CD:** Implemented COPPA-compliant parental consent flows, secure JWT authentication, Docker containerization, and automated GitHub Actions CI/CD deployment pipelines.

AI/ML ENGINEER INTERN (ENTERPRISE AI & RAG) | NIVESH AI

Jun 2026 – Present | Mumbai, India

- **Enterprise Speech AI Infrastructure:** Engineered scalable audio ingestion and processing microservices on Azure Cloud, supporting wealth-management intelligence for 10,000+ mutual fund distributors (MFDs).
- **Multilingual Speech-to-Text:** Integrated and optimized domain-specific STT models (Whisper, Sarvam AI) with spectral noise reduction (`noisereduce`), achieving high transcription fidelity on multilingual Indian English and Hindi audio.
- **Zero-Hallucination RAG Architecture:** Developed an enterprise RAG summarization system using ChromaDB vector search and attention-based LLMs to extract actionable portfolio reviews and next steps from advisory conversations.
- **Automated Data Extraction Pipelines:** Built multi-format document parsing pipelines (`pdfplumber`, `python-docx`) processing unstructured statements and audio transcripts into structured CRM records.
- **Measurable Business Impact:** Slashed advisor documentation turnaround time by 80%, automating manual note-taking and compliance tracking across tens of thousands of client interactions.
- **Cloud Reliability & Monitoring:** Containerized backend services using Docker on Azure, implemented asynchronous task queues, and established automated API health checks and structured logging.

FEATURED PROJECTS

VAAKSUDHI & VAAKIFY — PEDIATRIC SPEECH & DIAGNOSTIC MULTI-AGENT PLATFORM

Vaaksudhi Live · Vaakify Live · GitHub

Python, LangGraph, Faster-Whisper, Vosk, Kokoro ONNX, Librosa, Epitran, Q-Learning, FastAPI, React

- Architected an autonomous multi-agent clinical diagnostic platform evaluating speech errors across 8 Indian languages with sub-300ms phoneme alignment.
- Implemented LangGraph state machines and acoustic biofeedback algorithms to auto-generate objective clinical reports for speech therapists.
- Live in active clinical use with automated error taxonomy categorization and personalized practice games.

NIRVANA (FIREBOT) — AUTONOMOUS FIREFIGHTING ROBOT & SIMULATION

Live Console · GitHub · Docs

Python, MuJoCo Physics, PyTorch, Gymnasium, FastAPI, WebSockets, Three.js

- Built an autonomous firefighting robot operating inside a MuJoCo 3D physics simulator across multi-room environments with particle fire dynamics.
- Implemented deep reinforcement learning with multimodal sensor fusion (Lidar, IMU, thermal) for autonomous obstacle avoidance and fire source localization in <45s.
- Deployed a real-time web telemetry console on Render streaming robot state, sensor sweeps, and event logs via FastAPI and WebSockets.

TRIPWIRE — EDGE COMPUTER VISION VEHICLE COUNTER

Live Demo · GitHub

Python, OpenCV, FastAPI, React, Vite, WebSockets, Docker

- Engineered an edge-ready, CPU-only vehicle detection and tracking pipeline from live CCTV feeds with sub-50ms latency.
- Implemented Gaussian background subtraction, centroid tracking, and directional virtual tripwire algorithms on commodity CPUs without requiring GPUs.
- Shipped a low-latency monitoring dashboard streaming processed video over MJPEG and traffic crossing counts via WebSockets.

NYAYAI & AI BAIL RECKONER — LEGAL-REASONING LLM & RAG ASSISTANT

GitHub Repository

PyTorch, Hugging Face Transformers, ChromaDB, FastAPI, React

- Developed a domain-tuned legal AI assistant predicting bail hearing outcomes and citing judicial precedents across the Indian Penal Code (IPC) and Bharatiya Nyaya Sanhita (BNS).
- Implemented a dense RAG pipeline using ChromaDB vector embeddings to ground LLM reasoning strictly in verified Supreme Court rulings, eliminating hallucinations.
- Accelerated legal precedent retrieval from hours to seconds with verifiable case citations.

GECO-ENHANCED — UNSUPERVISED VISUAL SEMANTIC ALIGNMENT

GitHub Repository

PyTorch, DINOv2, Python Optimal Transport, FastAPI, React, Docker

- Built an unsupervised visual correspondence pipeline matching keypoints across different objects under large viewpoint and appearance shifts.
- Utilized Vision Transformer embeddings (DINOv2) and optimal transport algorithms for zero-shot cross-instance alignment without labeled training data.
- Built an interactive web tool for real-time feature visualization and semantic probe correspondence.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

SASTRA Deemed University, India

Expected Graduation: 2027

CERTIFICATIONS & RECOGNITIONS

- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems *(In-Person Practical Program)*
- **IIoT Certification:** Ministry of Heavy Industries (MHI) Program *(Project-Based Evaluation)*